

ICAD RT-4 · Set 9 — Answer Sheet

Exam 12-10-2026 · 75 Q · 300 marks · 3 hours · +4 correct, −1 wrong, 0 blank

Name

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Physics

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Numerical — write the value

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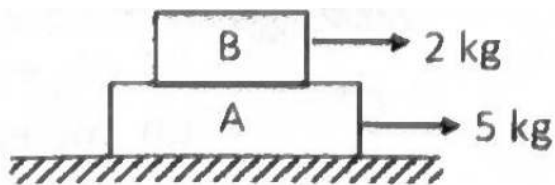
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Bubbles — single correct · Box — write the value · each subject out of 100

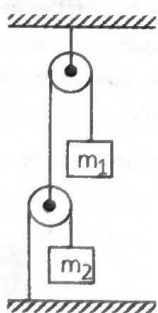
Physics

PART A • Q1–20 • SINGLE CORRECT

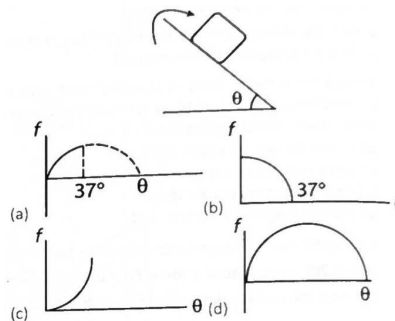
1. P2L4V3Q2 At $x = 5\pi/6$, $f(x) = 2\sin 3x + 3\cos 3x$ is: (o / max / min / none)
2. P2L6V6Q4 (Multiple correct) For $g'(x) = f(x)$ on $[a, b]$: $\int_a^b f(x)dx = g(b) - g(a)$; if $f \geq 0$ it equals geometric area; $\int_a^b = -\int_b^a$; average value vs $f(a)$
3. P3L6L7V4Q7 A and B are two vectors with angle θ between them. If $\sqrt{3}|A \times B| = A \cdot B$, the value of θ is: (a) 45 deg (b) 30 deg (c) 90 deg (d) 60 deg
4. P4L7V3Q2 A particle has $v_x = 5$ m/s and net velocity $v = 13$ m/s. Find v_y and then the slope: (a) 2.4 (b) 13 (c) 12 (d) 8
5. P4L8V4Q4 Ball thrown with speed v_0 at angle θ ; person runs at constant speed $v_0/2$ to catch it. Will he catch it? At what angle of projection? (a) 30 (b) 90 (c) 120 (d) 60
6. P4L8V5Q8 Ball projected from ground at angle θ ; after 1 s moving at 45° , after 2 s moving horizontally. Velocity of projection of the ball? (a) $10\sqrt{3} \text{ ms}^{-1}$ (b) $20\sqrt{3} \text{ ms}^{-1}$ (c) $10\sqrt{5} \text{ ms}^{-1}$ (d) $20\sqrt{2} \text{ ms}^{-1}$
7. P4L11V6Q10 Balloon rises with constant vertical velocity v_0 . Wind gives horizontal velocity component $v_x = ky$ (k constant, y = height risen). Find horizontal displacement from start point after ascending height h . (a) kh^2/v_0 (b) $kh^2/2v_0$ (c) $kh^2/3v_0$ (d) $2kh^2/v_0$
8. P4L12V2Q5 A boat with speed 5 km/hr in still water crosses a river of width 1 km along the shortest possible path in 15 minutes. Find the velocity of the river water (km/hr). (a) 1 (b) 3 (c) 4 (d) $\sqrt{41}$
9. P5L2V3Q1 Block B (2 kg) rests on block A (5 kg), both on the ground; the normal force between A and B is: (a) 20 N (b) 50 N (c) 30 N (d) 70 N



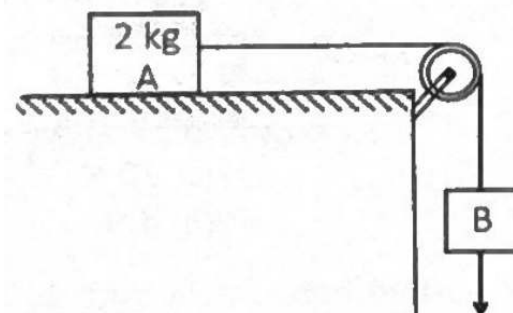
10. P6L3V5Q7 Acceleration of blocks m_1 and m_2 (pulleys massless/frictionless; strings inextensible; movable-pulley system). Options like $a_1 = (m_1 - 2m_2)g/(m_1 + 4m_2)$, $a_2 = 2(m_1 - 2m_2)g/(m_1 + 4m_2)$. (a) $a_1 = (m_1 - 2m_2)g/(m_1 + 4m_2)$, $a_2 = 2(m_1 - 2m_2)g/(m_1 + 4m_2)$ (b) $a_1 = (m_2 - 2m_1)g/(m_1 + 4m_2)$, $a_2 = 2(m_2 - 2m_1)g/(m_1 + 4m_2)$ (c) $a_1 = (m_2 - 2m_1)g/(m_2 + 4m_1)$, $a_2 = 2(m_2 - 2m_1)g/(m_2 + 4m_1)$ (d) $a_1 = (m_1 - 2m_2)g/(4m_1 + m_2)$, $a_2 = 2(m_1 - 2m_2)g/(4m_1 + m_2)$



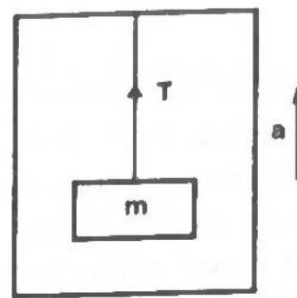
11. P7L1V6Q8 A block A of mass m is kept on a rough inclined plane of inclination θ . The coefficient of static friction between the block and the surface is $\mu_s = 0.75$. If the angle θ is slowly increased from 0° to 37° , then the friction versus θ graph is:



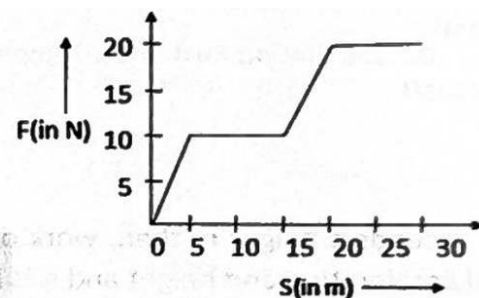
12. P7L4V3Q5 The coefficient of static friction μ_s between block A of mass 2 kg and the table is 0.2. A string from A passes over a smooth massless pulley at the table edge and carries block B. What is the maximum mass of B so that the two blocks do not move? ($g = 10 \text{ m/s}^2$) (a) 2.0 kg (b) 4.0 kg (c) 0.2 kg (d) 0.4 kg



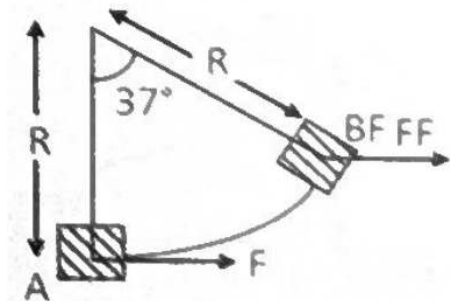
13. P8L1V4Q9 A block of mass m is suspended by a light thread from an elevator. The elevator starts from rest and accelerates upward with uniform acceleration 'a'. The work done during 't' second by the tension in the thread is: (a) $(m/2)(g + a)at^2$ (b) $(m/2)(g - a)at^2$ (c) $(m/2)gat^2$ (d) zero



14. P8L2V6Q7 A force acting on a body is as shown in the graph. The total work done in covering an initial distance of 20 m is: [Graph: F rises linearly $0 \rightarrow 10$ N over $s = 0 \rightarrow 5$ m; constant 10 N from 5 to 15 m; rises linearly $10 \rightarrow 20$ N from 15 to 20 m; then constant 20 N] (a) 225 J (b) 200 J (c) 400 J (d) 175 J



15. P8L3V3Q5 Two trucks, one loaded (A) and the other unloaded (B) are moving and have same kinetic energy. The mass of A is double that of B. Brakes are applied to both and are brought to rest. If distance covered by A before coming to rest is s_1 and that by B is s_2 , then: (a) $s_1 = s_2$ (b) $s_1 = 2s_2$ (c) $2s_1 = s_2$ (d) $s_1 = 4s_2$
16. P8L7V3Q5 A block of mass m is pulled along a circular arc (radius R , the arc subtending 37° at the centre) by means of a constant horizontal force F as shown. Work done by this force in pulling the block from A to B is: (a) $FR/2$ (b) $(3/5)FR$ (c) $(\sqrt{3}/2)FR$ (d) mgR



17. P4L9V4Q7 A ball is projected upward from a 70 m tower at 50 ms^{-1} , 30° with the horizontal. After how many seconds does it reach the ground? — (a) 2 s (b) 5 s (c) 7 s (d) 9 s
18. P5L3V5Q5 'A' and 'B' are two pegs on a vertical wall along a horizontal line, 13 cm apart. A body of mass 169 kg wt is suspended by a thread of 17 cm connected between A and B such that the two segments are perpendicular to each other. Find the tensions in the shorter and longer parts (write the value)
19. P8L3V4Q9 A 1000 kg elevator rises from rest in the basement to the fourth floor, a distance of 20 m. As it passes the fourth floor its speed is 4 m/sec. There is a constant frictional force of 500 N. The work done by the lifting mechanism is: (a) $196 \times 10^3 \text{ J}$ (b) $204 \times 10^3 \text{ J}$ (c) $214 \times 10^3 \text{ J}$ (d) $203 \times 10^5 \text{ J}$
20. P6L3V5Q10 (One or more correct) A system has a fixed pulley and a movable pulley. Block A hangs from the movable pulley; B at the free end. String length constant, pulleys massless. Position–time graph of A is concave upward. Which statements are correct? (write the value)

PART B · Q21–25 · NUMERICAL VALUE

21. P4L10V4Q4 A ball is thrown at speed v from level ground such that the area under its trajectory is maximum. If the angle of projection is π/k , find k .
22. P1L2V6Q2 Line $L: y = mx + c$ intersects axes at $(a, 0)$ and $(0, b)$, $a, b > 0$; if $a = 3$ and $c = 6$, determine b
23. P3L6L7V2Q4 Two vectors have magnitudes 2 m and 3 m; the angle between them is 60° . Find the scalar product of the two vectors.

24. P3L6L7V5Q7 A large number of particles are moving, each with velocity v having directions of motion randomly distributed. Find the average relative velocity between any two particles, averaged over all pairs.
25. P3L5V4Q1 A force $F = 3\hat{i} + c\hat{j} + 2\hat{k}$ acting on a particle causes a displacement $S = -4\hat{i} + 2\hat{j} - 3\hat{k}$ in its own direction. If the work done is 6J, find the value of c .

Chemistry

PART A · Q1–20 · SINGLE CORRECT

1. C2L2V5Q2 Beam $\nu = 5.0 \times 10^{14} \text{ Hz}$ delivers 10 J/s — photon energy & photons/s. (a) $E = 5.00 \times 10^{-15} \text{ J}$, $n = 2.00 \times 10^{12}$ photons/s (b) $E = 6.63 \times 10^{-34} \text{ J}$, $n = 1.50 \times 10^{31}$ photons/s (c) $E = 3.31 \times 10^{-19} \text{ J}$, $n = 3.02 \times 10^{19}$ photons/s (d) $E = 2.50 \times 10^{-19} \text{ J}$, $n = 4.00 \times 10^{19}$ photons/s
2. C2L5V4Q4 Energy of He^+ ion when its electron is in the 1st excited state. (a) -13.6 eV (b) -6.80 eV (c) -54.4 eV (d) -3.40 eV
3. C2L10V4Q1 In potassium, order of energy level for the 19th electron (4s vs 3d). (a) $3s > 3d$ (b) $4s < 3d$ (c) $4s > 4p$ (d) $4s = 3d$
4. C2L10V6Q8 Argon ($Z = 18$) has fully filled 2p and 3p subshells, filled in accordance with Hund's rule. Which of the following statements is/are correct? (a) Electrons in a p-subshell must occupy each orbital singly before pairing. (b) In a filled p-subshell, up-spin and down-spin electrons must always be equal. (c) The order in which electrons enter orbitals affects the final spin arrangement. (d) A filled p-subshell contains more up-spin electrons than down-spin electrons.
5. C3L9V3Q1 Normality of 24.5 g H_2SO_4 in 0.5 L (Eq. mass = 49). (a) 0.5 N (b) 1 N (c) 2 N (d) 3 N
6. C3L9V4Q5 1.0 g pure CaCO_3 needs 50 mL dilute HCl; strength of HCl. (a) 4 N (b) 2 N (c) 0.4 N (d) 0.2 N
7. C3L10V4Q5 A 5.2 molal aqueous solution of methyl alcohol, CH_3OH is supplied. What is the mole fraction of methyl alcohol in the solution? (0.086 / 0.050 / 0.100 / 0.190)
8. C3L10V6Q10 A 3 M sodium thiosulphate solution ($\text{Na}_2\text{S}_2\text{O}_3$) has a density of 1.25 g mL^{-1} . Which of the following statements regarding this solution are correct? (one or more options may be correct) (a) $\text{wt}\% \approx 37.92\%$ (b) mole fraction ≈ 0.065 (c) molality of $\text{Na}^+ \approx 7.72 \text{ m}$ (d) molality of $\text{S}_2\text{O}_3^{2-} \approx 3.86 \text{ m}$
9. C3L11V5Q4 In which of the following numbers are all zeros significant? (one or more options may be correct) (a) 5.0005 (b) 0.0030 (c) 30.000 (d) 0.5200
10. C4L3V4Q3 Among the following ions what is the atomic number having lowest ionic radii: Li^+ , Mg^{2+} , Al^{3+} (a) 3 (b) 12 (c) 13 (d) 11
11. C4L5V5Q4 Arrange N, O, P, S in increasing negativity (less negative $\rightarrow$ more negative) of electron gain enthalpy: (a) $\text{N} < \text{P} < \text{O} < \text{S}$ (b) $\text{N} < \text{O} < \text{P} < \text{S}$ (c) $\text{P} < \text{N} < \text{S} < \text{O}$ (d) $\text{N} < \text{O} < \text{S} < \text{P}$
12. C5L1V5Q8 Most favourable conditions for the formation of ionic bonds are (a) large cation and small anion (b) large cation and large anion (c) small cation and small anion (d) small cation and large anion
13. C5L3V4Q3 A π -bond is formed by the overlap of: (a) s-s orbitals (b) s-p orbitals (c) p-p orbitals in end to end fashion (d) p-p orbitals in sidewise manner

14. C5L5V5Q4 Assertion: The shape of XeF₄ is square planar. Reason: In an octahedral geometry, a single lone pair can occupy any position but a second lone pair will occupy the opposite position to the first. (a) If assertion is true but the reason is false. (b) If assertion is false but reason is true. (c) If both assertion and reason are true and the reason is the correct explanation of assertion. (d) If both assertion and reason are true but reason is not the correct explanation of assertion.
15. C5L6V3Q4 Determine shape and bond angle in NH₃. (a) Tetrahedral, 109.5° (b) Trigonal pyramidal, 107° (c) Bent, 104.5° (d) Linear, 180°
16. C5L6V3Q5 Identify geometry and bond angle of SF₆. (a) Octahedral, 90° (b) Trigonal bipyramidal, 120° (c) Tetrahedral, 109.5° (d) Square planar, 90°

PART B · Q21–25 · NUMERICAL VALUE

21. C1L4V6Q2 1.50 g gas over water 20 °C, 0.98 bar, VP 0.023 bar, 1000 mL; dry at NTP — volume. (a) 890 mL (b) 930 mL (c) 1000 mL (d) 760 mL
22. C2L7V4Q9 A certain transition in the hydrogen spectrum from an excited state to the ground state, in one or more steps, gives rise to a total of 10 lines. How many of these belong to the UV spectrum?
23. C1L1V3Q3 5 L container, 10 g gas, molar mass 20 — moles & volume. (a) 0.5 mol, 10 L/mol (b) 0.5 mol, 5 L/mol (c) 2 mol, 2.5 L/mol (d) 0.25 mol, 20 L/mol

Mathematics

PART A · Q1–20 · SINGLE CORRECT

1. M1L1V5Q4 A chord subtends an angle of θ radians at the centre of a circle of radius r . If the corresponding arc length equals twice the radius ($s = 2r$), what is θ , and in which quadrant does its terminal side lie if θ is taken as the principal representative in $(0, 2\pi)$? (a) $\theta = 2$, Quadrant I (b) $\theta = 2$, Quadrant II (c) $\theta = 2\pi/3$, Quadrant II (d) $\theta = 2\pi$, Quadrant IV
2. M2L2V5Q5 $A = \text{multiples of } 3, B = \text{multiples of } 5 \rightarrow A \cap B$. (a) {3, 6, 9} (b) {5, 10, 15, 20, ...} (c) {15, 30, 45, ...} (d) None of these
3. M2L3V4Q3 For $A, B, C \subseteq U$, $A \cap (B - C)$ equals (a) $(A \cap B) - C$ (b) $(A - B) \cap C$ (c) $(A - C) - B$ (d) $(A \cap B) - C'$
4. M3L2V4Q10 $(\sin 150^\circ - 5\cos 300^\circ + 7\tan 225^\circ)/(\tan 135^\circ + 3\sin 210^\circ)$. (a) 10 (b) -5 (c) -2 (d) -3/2
5. M3L3V6Q3 $2\cos^2 B - 1 = \tan^2 A \rightarrow \cos A \cdot \cos B$. (a) $\pm 1/2$ (b) $\pm 1/3$ (c) $\pm 1/\sqrt{2}$ (d) ± 1
6. M4L1V5SuggQ2 $A = \{a, e, i\}, B = \{2, 4, 6\}, f(a) = 2, f(e) = 4, f(i) = 6$ — domain & range
7. M4L2V6Q6 Even periodic f (period 10) piecewise — which correct? (multi). (a) $f(-4) = 40$ (b) $(f(-13) - f(11))/(f(13) + f(-11)) = 17/21$ (c) $f(5)$ is not defined (d) Range of $f(x)$ is $[0, 50]$
8. M5CL1V4Q7 $2A + B = \pi/2 \Rightarrow \sqrt{((1 + \sin B)/(1 - \sin B))} =$, $A \in (0, \pi/2)$ (a) $\tan A$ (b) $\cot A$ (c) $\tan B$ (d) $\cot B$
9. M5CL2V6Q3 A quadratic equation whose roots are $\tan 221/2^\circ$ and $\cot 221/2^\circ$ is (a) $x^2 - 2\sqrt{2}x + 1 = 0$ (b) $2x^2 - \sqrt{2}x + 1 = 0$ (c) $x^2 + 2\sqrt{2}x - 1 = 0$ (d) $x^2 - 2\sqrt{2}x - 1 = 0$
10. M5AL3V3Q1 $\cos(n+1)A \cos(n+2)A + \sin(n+1)A \sin(n+2)A = ?$ (a) $\cos nA$ (b) $\cos A$ (c) $\cos 2nA$ (d) $\cos 2A$

PART B · Q21–25 · NUMERICAL VALUE

21. M6L4V6Q3 If solution set of $x^2 - 5|x| + 6 > 0$ is $(-\infty, a) \cup (b, c) \cup (d, \infty)$ then value of $(|a| + |b| + |c| + |d|)$ is _____

17. C3L9V5Q2 0.63 g dibasic acid in 100 mL; 20 mL needs 10 mL N/5 NaOH — molecular mass. (a) 63 (b) 126 (c) 252 (d) 128
18. C3L12V5Q4 A sample of methane (CH₄) is burned completely in excess oxygen to form CO₂ and H₂O. If 16 g of CH₄ produces 44 g of CO₂, what mass of CO₂ would be produced from 4 g of CH₄ under similar conditions? (a) 8 g (b) 11 g (c) 22 g (d) 44 g
19. C5L5V6Q4 CF₄, SF₄ and XeF₄ contain the following electronic structure on their central atoms. Which one is the correct option? (a) 1, 2 and 3 lone pairs of electrons respectively (b) 0, 1 and 2 lone pairs of electrons respectively (c) 1, 1 and 1 lone pairs of electrons respectively (d) No lone pairs of electrons on any molecule
20. C1L9V5Q6 1 mol vdW gas $Z = PV/RT = 0.5$ at 0 °C & 100 atm, $b \approx 0$ — find a ($PV = RT - a/V$). (a) 0.63 L²·atm·mol⁻² (b) 1.256 L²·atm·mol⁻² (c) 2.50 L²·atm·mol⁻² (d) 3.00 L²·atm·mol⁻²
24. C1L11V4Q1 HCl polar but b.p. (-85 °C) far below HF (19.4 °C) — primarily because? (a) H–F bond is more covalent (b) HF molecules form strong hydrogen bonds (c) HF has larger molar mass (d) HCl molecules experience stronger dipole–dipole forces
25. C1L11V4Q9 Assertion: H₂O has a higher dielectric constant than H₂S. Reason: An extensive hydrogen-bond network in water increases molecular association and polarity. (a) Both A and R true and R explains A (b) Both true, R not explanation (c) A true, R false (d) A false, R true

11. M5CL4V3Q3 The maximum value of $\sin(x + \pi/6) + \cos(x + \pi/6)$ in the interval $(0, \pi/2)$ is attained at: (a) $x = \pi/12$ (b) $x = \pi/6$ (c) $x = \pi/3$ (d) $x = \pi/2$
12. M6L1V6Q1 If $a < b, a, b \in \mathbb{R}$ then the solution set of $(x - a)/(b - x) \geq 0$ is: (a) (a, b) (b) $[a, b)$ (c) $(-\infty, a] \cup (b, \infty)$ (d) $(-\infty, a] \cup [b, \infty)$
13. M8AL1V5Q5 If the n th terms of two A.P.s are $3n + 8$ and $7n + 15$, then the ratio of their 12th terms will be (a) 4/9 (b) 7/16 (c) 3/7 (d) 8/15
14. M8AL2V3Q4 If the sum of n terms of an A.P. be $3n^2 - n$, then its first terms is (a) 2 (b) 3 (c) 1 (d) 4
15. M8BL1V5Q5 The first and second terms of a GP are x^{-4} and x^n respectively. If x^{52} is the eight term of the same progression, then 'n' is equal to (a) 13 (b) 4 (c) 5 (d) 3
16. M8BL2V5Q7 If the sum of the series $1 + 2/x + 4/x^2 + 8/x^3 + \dots$ is a finite number, then (a) $x > 2$ (b) $x > -2$ (c) $x > 1/2$ (d) None
17. M2L3V5Q7 150 students who read at least one of 3 newspapers (65, 41, 50) → max reading all three. (a) 7 (b) 42 (c) 3 (d) Cannot be determined
18. M3L2V6Q8 $(x/a)\cos\theta + (y/b)\sin\theta = 1, (x/a)\sin\theta - (y/b)\cos\theta = 1 \rightarrow x^2/a^2 + y^2/b^2$. (a) 1 (b) -1 (c) 2 (d) 3
19. M4L3V5Q10 Number of onto functions $E = \{1, 2, 3, 4\} \rightarrow F = \{1, 2\}$ [IIT 2001]. (a) 14 (b) 16 (c) 12 (d) 8
20. M8CL1V4Q3 If a, b, c are three numbers which satisfy $a^2 + 4b^2 + c^2 = 4ab + 4bc - 2ac$, and if $a = 6, b = 10$, then the value of c is (write the value)
22. M6L5V3Q3 If Domain of $f(x) = 1/\sqrt{x - 1}$ is (a, ∞) then value of 'a' is:

- 23.** M7L1V3Q1 If $1/4^x = (400) \cdot 5^x$, then the value of x is:
- 24.** M8BL2V3Q4 If the common ratio of a G.P. is $2/3$ and sum up to the infinity is 12. Find first term.

- 25.** M5BL1V6Q2 If $(\cos 3A \cdot \sin 9A - \sin A \cos 5A) / (\cos A \cos 5A - \sin 3A \sin 9A) = \tan kA$, then $\sqrt{(k/2)}$ is equal to _____